



Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- *Dynamic learning environment that encourages problem-solving using emerging technology*
- *National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration*
- *Holistic, ethical and sustainable development for lifelong success.*

Session 2026-2027

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date: Krishna Mishra

14/07/26

(Signature and Date in Handwritten)

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Session	2026-27 (ODD)	Course Name	LAB: Lab Neural Networks and Deep Learning
Semester	7th	Course Code	23ADS1704
Roll No	41	Name of Student	Krishna Mishra

Practical Number	02
Course Outcome	1. Understand the fundamental concepts of deep learning and neural network architectures 2. Apply different techniques to develop and optimize deep learning models. 3. Analyze the performance of deep learning models using appropriate evaluation techniques. 4. Design and implement deep learning solutions for real-world problems using modern tools and frameworks.
Aim	Implement a Perceptron model using Google Colab.
Problem Definition	A university wants to automatically identify whether a flower belongs to the Setosa species or not based on its physical characteristics. Manual classification is time-consuming and prone to human error. Develop a Perceptron-based Machine Learning model using Google Colab that learns from the Iris dataset and predicts whether a flower is Setosa or Non Setosa based on sepal and petal measurements.
Theory (100 words)	The Perceptron is one of the simplest supervised machine learning algorithms used for binary classification. It learns a linear decision boundary by adjusting weights and bias based on the training data. During the learning process, the model iteratively updates its parameters to minimize classification errors. The Iris dataset is commonly used to demonstrate classification algorithms because it contains well-defined flower measurements. Since the Perceptron supports only binary classification, the original three-class dataset is converted into two classes: Setosa and Non-Setosa. The model's performance is evaluated using metrics such as accuracy, confusion matrix, precision, recall, and F1-score.

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Procedure and Execution

(100 Words)

Algorithm:

- ☐ Import the required Python libraries.
- ☐ Load the Iris dataset.
- ☐ Display the dataset for verification.
- ☐ Convert the multiclass target into binary classes (Setosa and Non-Setosa).
- ☐ Split the dataset into training and testing sets.
- ☐ Create the Perceptron model with suitable parameters.
- ☐ Train the model using the training data.
- ☐ Predict the class labels for the testing data.
- ☐ Calculate the model accuracy.
- ☐ Display the predicted values, actual values, confusion matrix, and classification report.

Code:

```
import numpy as np
import pandas as pd

from sklearn.datasets import load_iris
from sklearn.linear_model import Perceptron
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score,
confusion_matrix, classification_report
#load dataset
iris = load_iris()
df = pd.DataFrame(data=iris.data,
columns=iris.feature_names)
df['target'] = iris.target
print(df.head())

#variables
X= iris.data
y= iris.target
# Convert to binary classification
y = (y == 0).astype(int)

# Splitting dataset into train & test
X_train, X_test, y_train, y_test = train_test_split(
X,
y,
test_size=0.2,
random_state=42
)
# To create model
model = Perceptron (
    max_iter = 1000,
    eta0 = 0.1,
```



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```
random_state = 42
)
# To Train the model
model.fit(X_train, y_train)

# To make predictions
y_pred = model.predict(X_test)

#Accuracy of the model
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy:", accuracy)

# Predictions
print("\nPredicted values:")
print(y_pred)
print("\nActual values:")
print(y_test)

# Confusion Matrix
conf_matrix = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(conf_matrix)

# Confusion report
class_report = classification_report(y_test, y_pred)
print("\nClassification Report:")
print(class_report)
```



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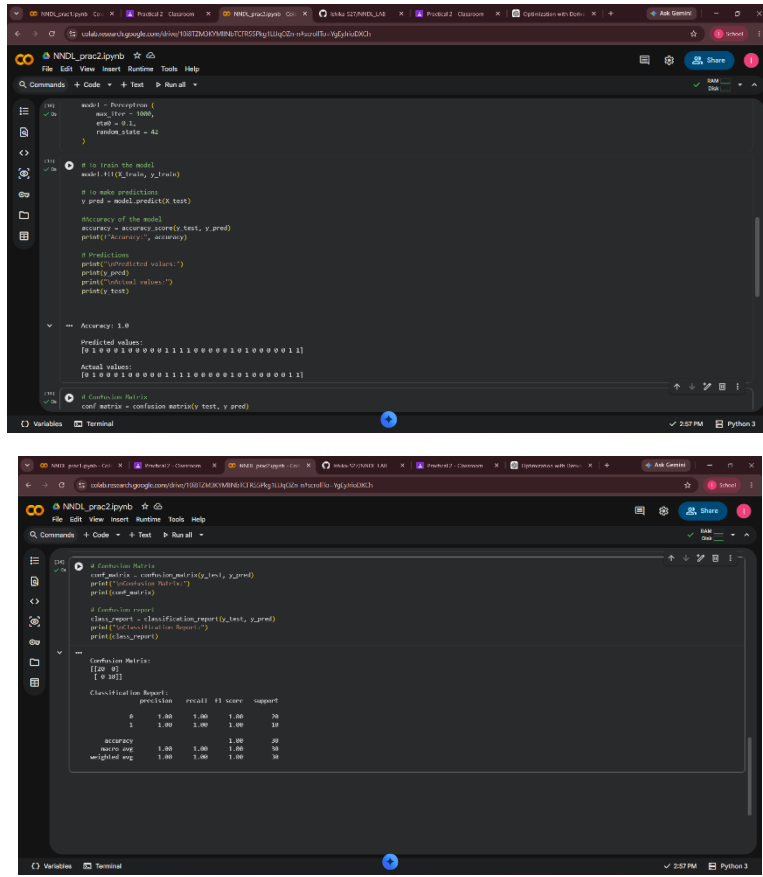
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	Output: 
Output Analysis	The Perceptron model successfully learns the relationship between flower measurements and their corresponding binary classes. After training, it predicts whether each test sample belongs to the Setosa or Non-Setosa category. The calculated accuracy indicates the percentage of correctly classified samples. The confusion matrix summarizes the numbers of correct and incorrect predictions, while the classification report provides precision, recall, F1-score, and support for each class. These evaluation metrics help assess the effectiveness and reliability of the trained classification model.
Link of student Github profile where lab assignment has been uploaded	krish-na45/Neural-Network-Deep-Learning-Practicals
Conclusion	This practical demonstrates the implementation of the Perceptron algorithm for binary classification using the Iris dataset. The experiment covers the complete machine learning workflow, including data loading, preprocessing,



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	model training, prediction, and evaluation. The obtained results show that the Perceptron can effectively classify Setosa and Non-Setosa flowers when the data is linearly separable. Overall, the practical provides a clear understanding of how a basic neural network model performs binary classification and how its performance is measured using standard evaluation metrics.																
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Date	08/07/26																